higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

13 April 2026

This marking guideline consists of 7 pages.

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TVET Exam Papers NATED & NCV	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	**N4** Mechanical Draughting NATED
N1-N3 Electrical Trade Theory NATED	**N2-N3** Diesel Trade Theory NATED	**N5-N6** Strength of Materials & Structures NATED	**N3** Building & Civil Technology NATED	**N1-N2** Fitting and Machining Theory NATED	**N1-N4** Plating and Structural Steel Drawing NATED
N1-N3 Motor Trade Theory NATED	**N5-N6** Mechanical Drawing & Design NATED	**N1-N2** Plumbing Theory NATED	**N4-N6** Fault Finding & Protective Devices NATED	**N2-N3** Instrument Trade Theory NATED	**N5-N6** Engineering Physics NATED
N5-N6 Fluid Mechanics NATED	**N4-N6** Supervisory Management NATED	**N4-N6** Digital Electronics NATED	**N6** Control Systems NATED	**N1-N2** Bricklaying &Plastering Theory NATED	More…

QUESTION 1

1.1
1.1.1 A
1.1.2 C
1.1.3 C
1.1.4 B
1.1.5 C

(5 × 1) (5)

1.2
1.2.1 True
1.2.2 False
1.2.3 False
1.2.4 True
1.2.5 True

(5 × 1) (5)
[10]

QUESTION 2

2.1 2.1.1

- 0 1 0 1 1, 1 0 1 →1/c→ 1* 10100,010✓
- 0 1 1 1 0, 1 1 0 →1/c→ 1* 10001,001✓

	1*	1	0	1	0	0,	0	1	0	
	1*	1	0	0	0	1,	0	0	1	
1	1*	0	0	1	0	1,	0	1	1	✓
↓	→	carry						+	1	
	1*	0	0	1	0	1,	1	0	0	✓

$1\,1\,0\,1\,0,0\,1\,1 = -26{,}375_{10}$ ✓ (5)

2.1.2 $42.6_{16} = 1000010{,}011_2$ ✓ & $44{,}25_{10} = 101100{,}010_2$ ✓

														1,	1	✓
1	0	1	1	0	0	0	1	1	0	0	0	0	1	0	0	1,	1	
								1	0	1	1	0	**0**	0	1	↓		
								0	1	0	1	1	0	0	0	1		✓
									1	0	1	1	0	0	0	1		
									0	0	0	0	0	0	0	0		✓

$\therefore 1{,}100_2 = 1{,}4_8$ ✓ (6)

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2.1.3 A, E_{16} = $1010,1110_2$✓ & $12,1_8$ = $1010,001_2$✓

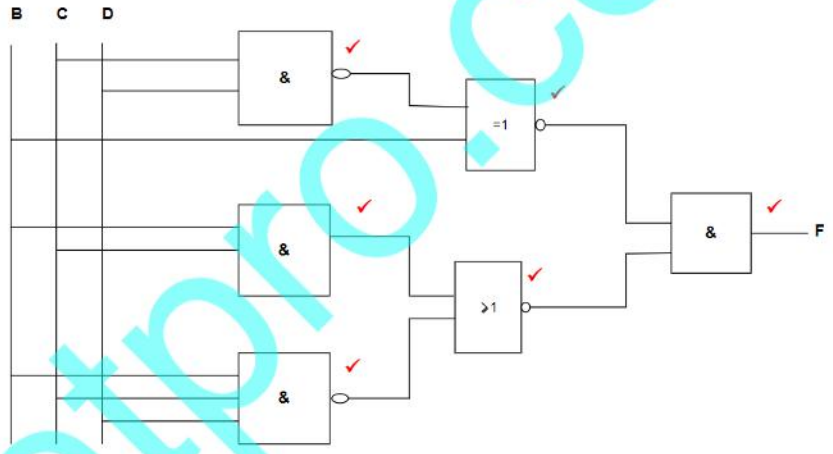
						1	0	1	0	1	1	1		
						1	0	1	0	0	0	1		
						1	0	1	0	1	1	1	✓	
			1	0	1	0	1	1	1					
1	0	1	0	1	1	1							✓	
1	1	0	1	1	1	0	0	0	0	1	1	1	✓	

$1\,1\,0\,1\,1\,1\,0, 0\,0\,0\,1\,1\,1_2 = 6\ E, 1\ C_{16}$✓ (6)

[23]

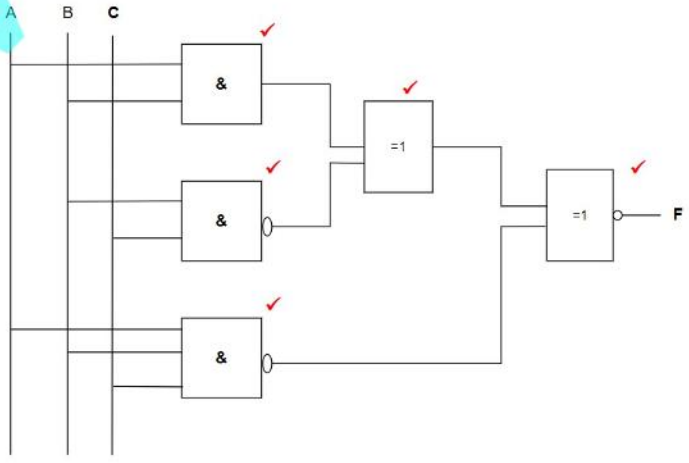
QUESTION 3

3.1 3.1.1 $\left(\overline{B \oplus \overline{CD}}\right)\left(\overline{BC + \overline{BCD}}\right) = F$



(6)

3.1.2 $\overline{\left(\overline{AB \oplus \overline{BC}}\right) \oplus \overline{ABC}} = F$

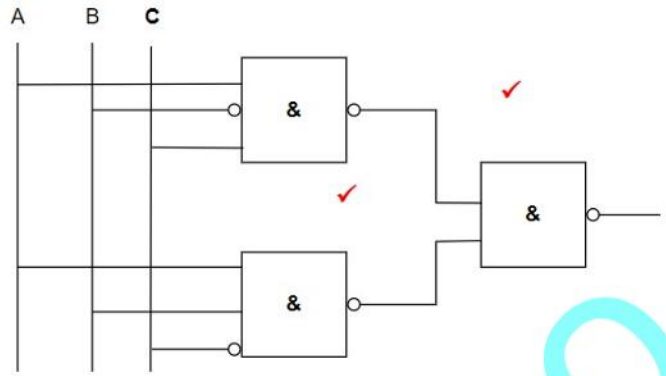


(5)

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3.2 3.2.1

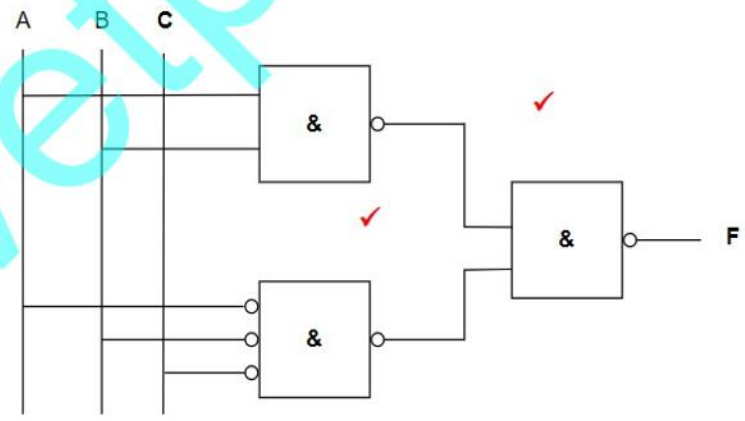
$$A(A+B+C)(\overline{B}+\overline{C})(\overline{A}+B+C)=F$$
$$=A+AB+AC(\overline{B}+\overline{C})(\overline{A}+B+C)\checkmark$$
$$=A(\overline{B}+\overline{C})(\overline{A}+B+C)$$
$$=(A\overline{B}+A\overline{C})(\overline{A}+B+C)\checkmark$$
$$=A\overline{B}C+AB\overline{C}\checkmark$$
$$=\overline{\overline{A\overline{B}C+AB\overline{C}}}\checkmark$$
$$=\overline{\overline{A\overline{B}C}.\overline{AB\overline{C}}}\checkmark$$



(7)

3.2.2

$$AB+\overline{\overline{A}B\overline{C}+A\overline{B}+\overline{A}C}=F$$
$$=AB+(\overline{\overline{A}B\overline{C}}.\overline{A\overline{B}}.\overline{\overline{A}C})$$
$$=AB+(A+\overline{B}+C)(\overline{A}+B)(A+\overline{C})\checkmark$$
$$=AB+(AB+\overline{A}\ \overline{B}+\overline{A}C+BC)(A+\overline{C})\ \checkmark$$
$$=AB+AB+ABC+AB\overline{C}+\overline{A}\ \overline{B}\ \overline{C}\checkmark$$
$$=AB+\overline{A}\ \overline{B}\ \overline{C}\checkmark$$
$$=\overline{\overline{AB+\overline{A}\ \overline{B}\ \overline{C}}}$$
$$=\overline{\overline{AB}.\overline{\overline{A}\ \overline{B}\ \overline{C}}}\checkmark$$

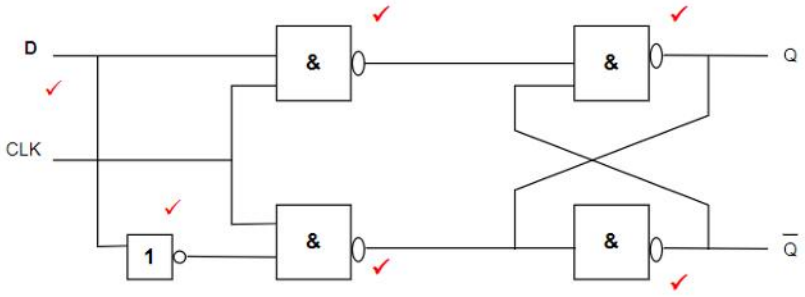


(7)
[25]

QUESTION 4

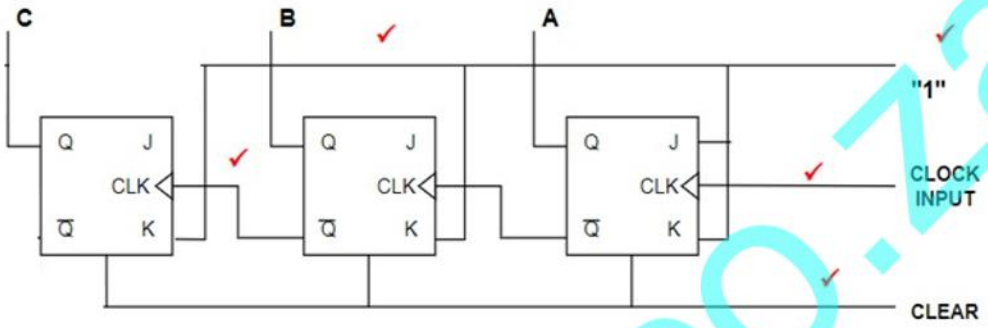
4.1 Gated equivalent of a clocked D latch

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(6)

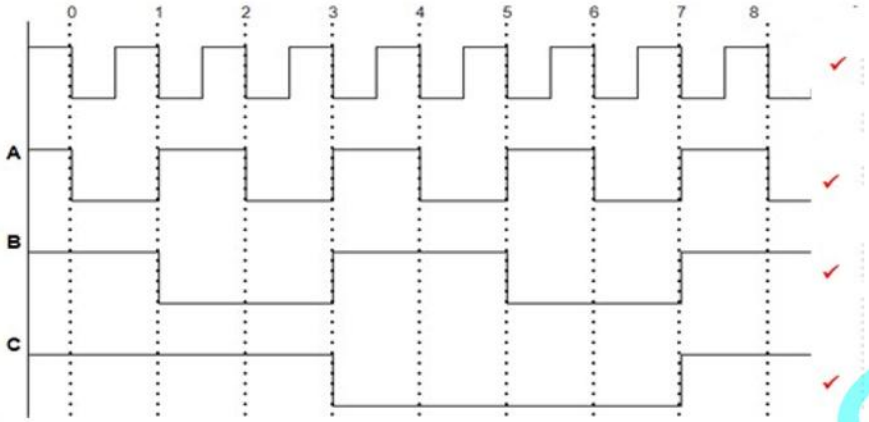
4.2 3-bit asynchronous down counter



Truth table

Decimal	Pulses	C ✓	B ✓	A ✓
7	0	1	1	1
6	1	1	1	0
5	2	1	0	1
4	3	1	0	0
3	4	0	1	1
2	5	0	1	0
1	6	0	0	1
0	7	0	0	0
7	8	1	1	1

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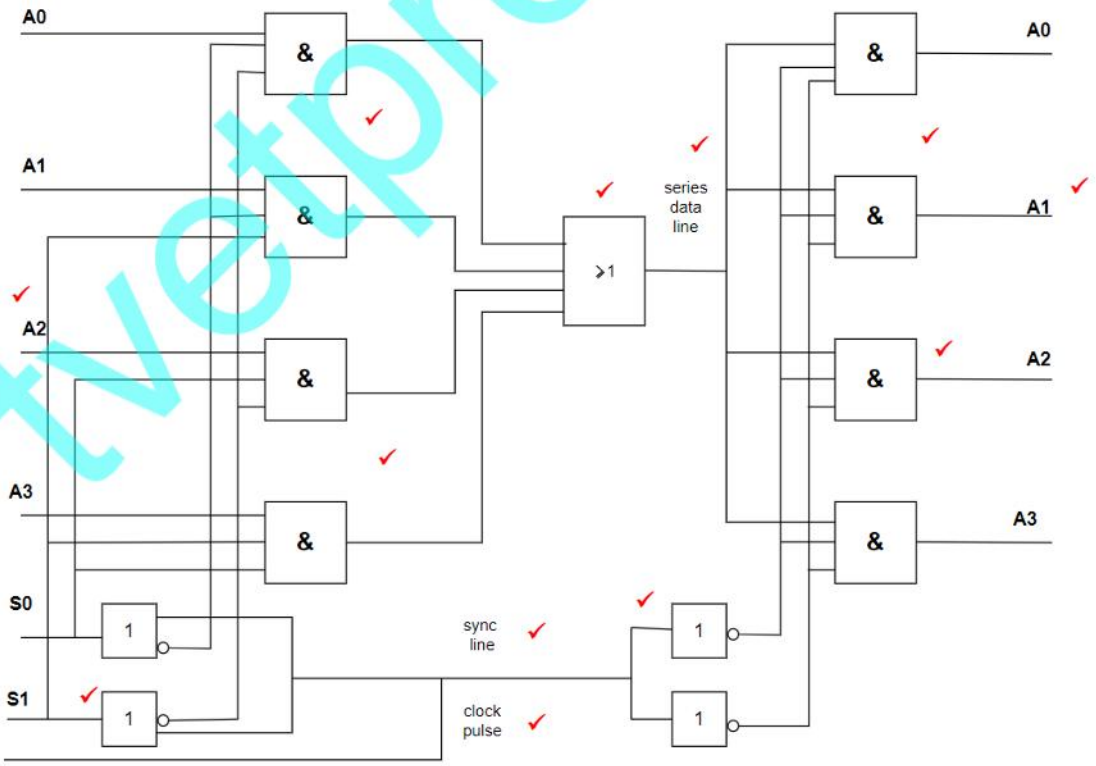
Up to 8 clock pulse: follow the sequence as per truth table. (12)

4.3
- Fan in is the number of inputs available to be coupled to a specific logic output.✓✓
- Fan out is the maximum number of standard inputs that can reliably be driven by one output.✓✓ (2 × 2) (4)

[22]

QUESTION 5

5.1 4:1 Multiplexer/Demultiplexer



(12)

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5.2
- The reference voltage V, is generated in steps by the D/A converter.✓
- The control unit resets the digital counter and D/A converter to zero.✓
- The start pulse opens the gates so that repetitive pulses form the clock pulse generator can enter the counter and D/A converter.✓✓
- The step voltage v, increases and as soon as Vr = Vm the polarity of the comparator changes which sends a stop to the clocking gate.✓✓
- The counter stops and the count, that is equivalent to the measured voltage will be displayed.✓
- The feedback loop of D/A converter is opened only at the moment that Vr > Vm.✓

(8)
[20]

TOTAL: 100

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